Practice Set 18 SAQ Sheet – Short Answer Practice

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Short-answer practice sheet compiled from handwritten study lists.

Q3a. Fungal UTI

Type: SAQ (10 marks)

Introduction and organisms

- Fungal UTI (urinary tract infection) is rising with antibiotics, catheters and immunosuppression. **Candida** is commonest: *C. albicans*, then *C. glabrata* (often fluconazole-resistant); *C. krusei* is intrinsically resistant.
- Others in AIDS (acquired immunodeficiency syndrome) or transplant patients: Aspergillus, Cryptococcus, Mucorales, Histoplasma.

Risk factors and routes

- Antibiotics, indwelling catheters and stents, diabetes, steroids, neutropenia, prematurity, parenteral nutrition, obstruction.
- **Ascending** (catheter) causes most candiduria. **Haematogenous** spread from candidaemia: in disseminated candidiasis the kidney is the organ most often involved.

Clinical spectrum

- Asymptomatic candiduria (most) → cystitis → pyelonephritis → **fungus ball (bezoar)** causing obstruction → renal abscess or candidaemia.

Diagnosis

- Microscopy: budding yeasts and pseudohyphae (may mimic red cells). Repeat culture: contamination is common.
- Children: treat when repeated cultures exceed **10,000–15,000 CFU (colony-forming units)/mL**.
- Persistent funguria: ultrasound or CT (computed tomography) for bezoar; blood cultures if high risk.

Management

Situation	Treatment
Asymptomatic	Remove or change catheter, stop antibiotics; no antifungal
Treat asymptomatic only if	Neutropenia, very-low-birth-weight infant, before a urological procedure
Cystitis	Fluconazole 200 mg (3 mg/kg) daily for 2 weeks
Pyelonephritis	Fluconazole 200–400 mg (3–6 mg/kg) daily for 2 weeks
Fluconazole-resistant (*C. glabrata*, *C. krusei*)	Amphotericin B deoxycholate 0.3–0.6 mg/kg/day for 1–7 days ± flucytosine
Resistant cystitis	Amphotericin B bladder irrigation 50 mg/L
Fungus ball	PCN (percutaneous nephrostomy) drainage + systemic antifungal + local amphotericin irrigation; endoscopic or open removal if persistent

- Fluconazole concentrates in urine; **echinocandins and lipid amphotericin do not.**
- **Before procedures:** IDSA (Infectious Diseases Society of America) 2016 treats all candiduria (fluconazole 400 mg daily for several days before and after). AUA (American Urological Association) Best Practice Statement 2019 gives single-dose antifungal prophylaxis only for urothelium-breaching surgery. The two disagree.
- **Also know:** alkalinisation for bezoar; fluconazole caution under 6 months.

Diagram to draw: budding yeast with pseudohyphae; fungus ball in renal pelvis with nephrostomy.

Key points to remember:

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.
- Fungus ball: drain, antifungal, remove.

Q3b. Antimicrobial resistance

Type: SAQ (10 marks) (also answered in part in 2015-Summer Paper 1, Q3f)

Definition

- AMR (antimicrobial resistance): a microbe grows despite drug levels achievable in the patient. Reported as S (sensitive), I (intermediate), R (resistant) with the MIC (minimum inhibitory concentration).
- One of the most serious global health threats.

Types

Type	Mechanism	Relevance
Intrinsic	Species lacks the target	Predictable (enterococci and cephalosporins)
Acquired chromosomal	Mutants selected when urine level is close to MIC	Underdosing, poor compliance
Plasmid (extrachromosomal)	Horizontal transfer in bowel flora	**Commonest**; multidrug resistance after antibiotics

Mechanisms

- **Enzymes:** β-lactamases; aminoglycoside-modifying enzymes.
- **Altered target:** penicillin-binding protein; DNA gyrase mutation (quinolones).
- **Reduced entry** (porin loss) and **efflux pumps**.
- **Bypass:** enterococci scavenge folate (escape TMP-SMX (trimethoprim–sulfamethoxazole)).

Key resistant organisms

- **ESBL (extended-spectrum β-lactamase)** (plasmid, 1983): resists 3rd-generation cephalosporins; often co-resistant to fluoroquinolones and aminoglycosides. *E. coli* ST131 carries it.
- **AmpC; KPC (*Klebsiella pneumoniae* carbapenemase, 2001); NDM (New Delhi metallo-β-lactamase)**: carbapenem resistance.
- MRSA (methicillin-resistant *S. aureus*), VRE (vancomycin-resistant enterococci).

Drivers in urology

- Overuse of fluoroquinolones and cephalosporins; treating ASB (asymptomatic bacteriuria); long prophylaxis; catheters, stents, stones (biofilm).
- Transrectal biopsy: fluoroquinolone-resistant rectal *E. coli* causes sepsis.

Prevention: stewardship

1. Culture before treatment; follow the **local antibiogram**.
2. TMP-SMX only if local resistance ≤20%; oral fluoroquinolone for pyelonephritis only if ≤10%.
3. Prefer narrow, urine-confined drugs for cystitis (nitrofurantoin, fosfomycin, pivmecillinam: low resistance).
4. Shortest course; de-escalate to culture.
5. Do not treat ASB, except in pregnancy and before mucosa-breaching procedures.
6. Single-dose surgical prophylaxis; transperineal biopsy or rectal-swab-targeted prophylaxis.
7. Infection control and hand hygiene.

- **Also know:** FDA (Food and Drug Administration) 2016 fluoroquinolone warning; *C. difficile*; ESBL: carbapenem, or fosfomycin/nitrofurantoin for cystitis.

Recent advances

- New β-lactam/β-lactamase inhibitor combinations (ceftazidime–avibactam, meropenem–vaborbactam); cefiderocol; gepotidacin.

Diagram to draw: Gram-negative cell showing β-lactamase, porin loss, efflux pump, altered target.

> **Key points to remember:**
> - Plasmid-mediated resistance is commonest.
> - ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
> - TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
> - Stewardship: culture, narrow, short, no ASB treatment.

Q3c. Biofilm

Type: SAQ (10 marks) (also answered in part in 2015-Summer Paper 3, Q3d)

Definition

- A **structured microbial community in a self-made polymeric matrix**, adherent to a living or inert surface. The matrix contains exopolysaccharide, extracellular DNA and proteins.

Stages of formation

1. **Conditioning film:** host proteins coat the device within minutes.
2. **Attachment:** reversible, then irreversible (pili, adhesins, flagella).
3. **Matrix production and microcolonies.**
4. **Maturation:** 3-D (three-dimensional) structure with water channels; cell-to-cell signalling (quorum sensing).
5. **Dispersion:** free bacteria seed new infection.

Why antibiotics and immunity fail

- Poor drug penetration through the matrix.
- Slow-growing, dormant cells resist drugs that need active growth.
- Altered genes, envelope and targets.
- Survive **1000–1500 times** the usual killing concentration. Neutrophils are excluded.

Urological examples

Site	Point
Urethral catheter	Forms within minutes; **the catheter, not the bladder, is infected**. Proteus urease makes crystalline biofilm → encrustation and blockage
Ureteric stent, nephrostomy	Stent colonised more often than urine shows; nephrostomy changed every 3 months
Penile prosthesis, AUS (artificial urinary sphincter)	Revision of an "uninfected" device: about 10% infection
Infection stones	Bacteria inside struvite; antibiotics cannot sterilise
Chronic bacterial prostatitis	Biofilm commoner in prostatitis *E. coli*; needs longer, penetrating therapy
Recurrent cystitis	**Intracellular bacterial communities**: biofilm-like pods inside umbrella cells

- Common organisms: Pseudomonas, Proteus, *E. coli*, Enterococcus, coagulase-negative staphylococci, Candida.

Prevention and management

- Avoid devices; shortest dwell; aseptic insertion; closed dependent drainage.
- **Change the catheter** when treating symptomatic CAUTI (catheter-associated UTI (urinary tract infection)); remove infected stent or implant.
- Silver-alloy catheters reduce bacteriuria, not clinical UTI.
- Implants: antibiotic-coated devices, no-touch technique, **antiseptic washout** at revision (infection about 10% → 3%).
- Stones: complete clearance. Chemodissolution for catheter encrustation. Macrolides penetrate biofilm (prostatitis).
- **Also know (research):** quorum-sensing inhibitors, bacteriophages, new coatings.

Diagram to draw: five-stage biofilm cycle on a catheter surface.

Key points to remember:

- Biofilm: microbes in self-made matrix on a surface.
- Survives 1000–1500 times normal antibiotic levels.
- Treat CAUTI by changing the catheter.
- Revision washout cuts prosthesis infection from about 10% to 3%.

Q3d. Tumour markers in urology

Type: SAQ (10 marks)

Definition and uses

- A substance made by a tumour, measured in blood, urine or tissue.
- Uses: screening, diagnosis, staging, prognosis, response, and detecting relapse. Ideal: specific, sensitive, proportional to bulk, short half-life.

Prostate

- **PSA (prostate-specific antigen):** a kallikrein serine protease (hK3) from luminal cells; **organ-specific, not cancer-specific** (BPH (benign prostatic hyperplasia), prostatitis, instrumentation raise it). Best marker for post-treatment follow-up.
- **Refinements:** density (PSA ÷ volume; cut-offs 0.08–0.15); %free PSA (lower in cancer; for PSA 4–10 ng/mL); velocity (little value); **PHI (Prostate Health Index) = (p2PSA ÷ free PSA) × √total PSA**; 4Kscore.
- **Urine:** PCA3 (prostate cancer antigen 3) after DRE (digital rectal examination); score = PCA3 mRNA ÷ KLK3 mRNA × 1000.
- **Also know:** tissue genomic classifiers.

Testis (germ cell tumours)

Marker	Half-life	Source	Never raised in
AFP (α-fetoprotein)	5–7 days	Embryonal, yolk sac	Pure seminoma, choriocarcinoma
β-hCG (human chorionic gonadotrophin)	24–36 hours	Choriocarcinoma, embryonal; seminoma in ~15%	Yolk sac tumour
LDH (lactate dehydrogenase)	24 hours	Non-specific; reflects bulk	—

- Measure before and after orchidectomy, on chemotherapy, in follow-up. Post-orchidectomy values give the **S stage** (S1: LDH <1.5 × normal, hCG <5000 mIU/mL, AFP <1000 ng/mL) and the **IGCCCG (International Germ Cell Cancer Collaborative Group)** risk group.
- Raised AFP with "pure seminoma" histology → treat as NSGCT (non-seminomatous germ cell tumour). False positives: liver disease, marijuana, LH (luteinising hormone) cross-reaction.
- **miR-371a-3p:** sensitive new marker; not raised by teratoma.

Bladder and upper tract

- **Urine cytology:** sensitive and specific for high-grade cancer, poor for low grade; for surveillance, not screening.
- NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), ImmunoCyt, UroVysion FISH (fluorescence in situ hybridisation): selected use only.

Others

- **Kidney and penis:** no validated serum marker.
- **Adrenal:** metanephrines, cortisol, aldosterone–renin ratio. **Neuroblastoma:** urinary VMA (vanillylmandelic acid)/HVA (homovanillic acid) raised in 90–95%.

Diagram to draw: marker decline curve after orchidectomy using half-lives.

Key points to remember:

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

Q3e. Haemostatic agents in urology

Type: SAQ (10 marks)

Introduction

- Topical agents are **adjuncts** to surgical haemostasis (suture, clip, energy), used for diffuse ooze. No randomised study shows clear benefit of any single agent in partial nephrectomy.

Classification

Class	Examples	Mechanism / note
Passive (mechanical)	Gelatin sponge (Gelfoam), **oxidised regenerated cellulose (Surgicel)**, microfibrillar collagen, polysaccharide spheres	Matrix for platelets; needs intact clotting. Cellulose is acidic (antibacterial)
Active	Topical thrombin	Converts the patient's fibrinogen to fibrin
Flowable	Gelatin matrix + thrombin (**Floseal**)	Fills irregular beds, swells
Fibrin sealants	Fibrinogen + thrombin (Tisseel, Evicel)	Final clotting step; works despite coagulopathy; also seals leaks
Synthetic sealants and glues	PEG (polyethylene glycol) hydrogel (Coseal), cyanoacrylate, albumin–glutaraldehyde (BioGlue)	Seal; adhesive
Systemic	Tranexamic acid, aminocaproic acid, desmopressin	Antifibrinolytic; desmopressin for uraemic platelets

Urological uses

- **Partial nephrectomy:** renorrhaphy ± sealant or Surgicel bolster; sutureless repair in selected cases.
- **Laparoscopic venous injury:** pressure → sealant → suture.
- PCNL (percutaneous nephrolithotomy) tract sealing in tubeless cases; adrenal and spleen/liver ooze; prostatectomy and lymphadenectomy beds.
- **Fibrin glue:** ureteroscopic fulguration for upper-tract bleeding; fistula plugs.
- **Tranexamic acid:** lowers transfusion at radical cystectomy.

Intravesical agents (haemorrhagic cystitis)

Agent	Note
Alum 1% (50 g in 5 L)	First line; avoid in renal failure (aluminium toxicity)
Silver nitrate 0.5–1%	Chemical coagulation
Aminocaproic acid 200 mg/L irrigation	Thromboembolism and clot retention
Formalin 1–4%	Most effective; under anaesthesia; **cystogram first** to exclude reflux

Complications

- Swelling compresses nerves, ureter or spinal cord (gelatin, cellulose): use the least amount.
- Mimics abscess on imaging; foreign-body reaction; infection.
- Flowables entering open vessels → embolism or thrombosis.
- Bovine thrombin → antibodies and coagulopathy; human products → theoretical viral risk.

- **Also know:** energy adjuncts (bipolar, argon beam, vessel sealers); hyperbaric oxygen in radiation cystitis.

Diagram to draw: clotting cascade showing where passive agents, thrombin and fibrin sealant act.

> **Key points to remember:**
>
> - Agents are adjuncts, not substitutes for sutures.
> - Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
> - Surgicel: acidic, may mimic abscess.
> - Alum first line; formalin last, after cystogram.

Q3f. Urethral catheters, guidewires and DJ stents

Type: SAQ (10 marks) (also answered in part in 2016-Summer Paper 1, Q3e and 2015-Summer Paper 3, Q1)

Urethral catheters

- **Size:** Fr (French) = 3 × diameter in mm. Adult 14–16 Fr; clot retention 20–24 Fr three-way. Male meatus is the narrowest point (24 Fr).
- **Lumens:** one (intermittent, specimen); two (drainage + balloon, routine); three (adds irrigation; stiff wall resists collapse). Each lumen shrinks drainage.
- **Tips:** straight; **coudé** (curved, for prostatic enlargement, tip up); **Councill** (open tip, over a wire).
- **Materials:** latex, silicone (better urethral tolerance), PVC (polyvinyl chloride); silver coatings reduce bacteriuria, not clinical infection.
- **Special:** Malecot (winged, no balloon); Lapides diagnostic (rings 1 cm apart); urodynamic.
- Inflate balloon with sterile water, only after urine flows at the hub. **Suspected urethral injury → RUG (retrograde urethrogram) first.**

Guidewires

Feature	Details
Uses	Access, safety wire, straightening ureter, guiding dilation and stents
Construction	Stainless-steel core with outer wrap; **nitinol** core is kink-resistant
Coating	PTFE (polytetrafluoroethylene): standard safety wire. **Hydrophilic** "glide": passes impacted stones, but too slippery to be a safety wire. Hybrid: hydrophilic tip, PTFE shaft
Size	0.018–0.038 inch (commonly 0.035/0.038); 80–260 cm, 145 cm most useful; floppy tip 1–3 cm
Tip and stiffness	Straight, angled, J; standard to super-stiff (straightens tortuous ureter)

DJ (double-J) stents

- Self-retaining double-pigtail (Finney, 1978); adult usually 6 Fr, length by height.
- **Materials:** polyurethane, silicone (hydrocoated silicone less painful); **metallic (Resonance)** for extrinsic malignant compression.
- **Indications:** obstruction (infected obstructed kidney, malignancy, pregnancy), after URS (ureteroscopy) with injury, pyeloplasty, transplant, leak.
- **Placement:** retrograde pyelogram → wire → stent over wire under fluoroscopy or vision.
- **Problems:** stent symptoms (bladder irritation and reflux) eased by α-blockers and antimuscarinics; migration; encrustation (forgotten stent); failure with extrinsic compression.

Diagram to draw: three-way Foley cross-section; DJ stent in situ.

Key points to remember:

- Fr = 3 × diameter (mm); coudé for prostatic enlargement, Councill over a wire.
- Hydrophilic wire to pass, PTFE wire to stay (safety).
- DJ symptoms: α-blockers and antimuscarinics help.
- Long-dwelling stents: monitor for encrustation.

Rapid Revision Sheet – Practice Set 18 SAQ Sheet

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q3a. Fungal UTI (SAQ)

- Candida commonest; *C. glabrata* often fluconazole-resistant.
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